GROUP WORK PROJECT # ___
GROUP NUMBER: ____________

MScFE 622: Stochastic Modeling

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Group #: 739

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MScFE 622 — Stochastic Modeling

Group Work Project #2: HMM Regime Allocation

Group member
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Analytical scope

This notebook is the canonical technical artifact for the HMM-only revision. The only regime model used in Steps 2–5 is a **Gaussian Hidden Markov Model (HMM)** fitted to daily changes in the VIX. We evaluate exactly two state counts, $K = 2$ and $K = 3$, select between valid candidates by BIC, construct both assignment allocation rules (**100% Keep** and **60/40 Spread**), and compare both HMM strategies with the required monthly equal-weight and SPY buy-and-hold benchmarks.

All numerical tables and figures below are rendered from committed canonical artifacts generated from `data/processed/step1_data.csv`. Analytical calculations are kept outside the notebook so that the notebook remains a presentation and orchestration layer.

In [1]:
```
from vix_regime_allocation import notebook_helpers as nb
```

In [2]:
```
from vix_regime_allocation import notebook_sensitivity as sensitivity_nb
```

Step 1 — Canonical input sample

The revision does not re-download data. The common input is the committed Step 1 dataset containing TLT, GLD, SPY, VIX, their ETF log returns, and the daily VIX change. All later model fitting, state statistics, allocations, backtests, and sensitivity calculations use this same date-indexed sample.

VIX is an options-implied measure designed to summarize the market's expectation of near-term S&P 500 volatility, so changes in VIX provide a natural single observation series for a volatility-regime model (Whaley 98–105).

In [3]:
```
nb.step_1_data_overview()
```

Step 1 — Common sample and transformed series

Date	TLT	GLD	SPY	VIX	TLT_log_return	GLD_log_return	S
2004-11-19	43.302605	44.779999	78.858780	13.50	-0.008012	0.008973	
2004-11-22	43.528297	44.950001	79.234833	12.97	0.005198	0.003789	
2004-11-23	43.582287	44.750000	79.355743	12.67	0.001240	-0.004459	
2004-11-24	43.582287	45.049999	79.543793	12.72	0.000000	0.006682	
2004-11-26	43.297714	45.290001	79.483330	12.78	-0.006551	0.005313	

	count	mean	std	min	25%	50%	
TLT_log_return	5465.0	0.000114	0.009156	-0.069011	-0.005299	0.000432	0.0
GLD_log_return	5465.0	0.000405	0.011492	-0.108412	-0.005154	0.000590	0.0
SPY_log_return	5465.0	0.000416	0.011904	-0.115887	-0.003935	0.000718	0.0
VIX_change	5465.0	0.000404	1.901620	-18.710003	-0.720000	-0.099999	0.5

Out[3]:

Date	TLT	GLD	SPY	VIX	TLT_log_return	GLD_log_retur
2004-11-19	43.302605	44.779999	78.858780	13.50	-0.008012	0.00897
2004-11-22	43.528297	44.950001	79.234833	12.97	0.005198	0.00378
2004-11-23	43.582287	44.750000	79.355743	12.67	0.001240	-0.00445
2004-11-24	43.582287	45.049999	79.543793	12.72	0.000000	0.00668
2004-11-26	43.297714	45.290001	79.483330	12.78	-0.006551	0.00531
...	...	...	...	...	...	.
2026-08-11	82.190002	400.959991	770.559998	15.28	0.001583	-0.00393
2026-08-12	82.110001	404.920013	772.489990	14.55	-0.000974	0.00982
2026-08-13	82.589996	398.959991	777.880005	14.63	0.005829	-0.01482
2026-08-14	82.040001	401.480011	776.340027	14.25	-0.006682	0.00629
2026-08-17	81.349998	405.489990	772.669983	15.19	-0.008446	0.00993

5465 rows × 8 columns

Step 2 — Gaussian Hidden Markov Models

2.1 Latent-state model

An HMM assumes that the observed series is generated by an **unobserved first-order state process**. Let S_t be the latent state on date t and X_t the observed daily VIX change. Conditional on S_t, the observation distribution is Gaussian; dependence across dates enters through state-transition probabilities. This separates persistent latent regimes from noisy observed movements (Rabiner 257–286).

Greek letters used in the next equation

- π — **pi** — initial-state probability.

- μ — **mu** — state-conditional mean.
- σ — **sigma** — state-conditional standard deviation.

$$P(S_1 = j) = \pi_j, \qquad P(S_t = j \mid S_{t-1} = i) = a_{ij}, \qquad X_t \mid S_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2).$$

The parameter set therefore consists of initial-state probabilities, a row-stochastic transition matrix, state-specific means, and state-specific variances. In this project the emission covariance is diagonal, the observation is one-dimensional `VIX_change`, and candidate state counts are exactly $K = 2$ and $K = 3$.

2.2 Observed-data likelihood

The latent state path is not observed, so the likelihood sums over every possible state sequence. The forward recursion evaluates this sum efficiently rather than enumerating all paths (Rabiner 257–286).

$$L = \sum_{s_1,\ldots,s_T} P(S_1 = s_1) \prod_{t=2}^{T} P(S_t = s_t \mid S_{t-1} = s_{t-1}) \prod_{t=1}^{T} f(X_t \mid S_t = s_t).$$

2.3 Expectation-Maximization / Baum-Welch estimation

Baum-Welch is the HMM-specific Expectation-Maximization procedure. The **E-step** uses forward-backward probabilities to compute posterior responsibilities for individual states and adjacent state transitions. The **M-step** re-estimates initial probabilities, transition probabilities, Gaussian means, and variances from those responsibilities. The two steps repeat until the configured convergence criterion is met (Baum et al. 164–171; Rabiner 257–286).

Greek letters used in the next equation

- γ — **gamma** — posterior probability that one state is occupied at a date.
- ξ — **xi** — posterior probability of one adjacent state transition.

$$\gamma_t(j) = P(S_t = j \mid X_{1:T}), \qquad \xi_t(i,j) = P(S_t = i, S_{t+1} = j \mid X_{1:T}).$$

Greek letters used in the next equation

- π — **pi** — initial-state probability.
- μ — **mu** — state-conditional mean.
- σ — **sigma** — state-conditional standard deviation.
- γ — **gamma** — state responsibility.
- ξ — **xi** — transition responsibility.

$$\pi_j^{new} = \gamma_1(j), \qquad a_{ij}^{new} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \qquad \mu_j^{new} = \frac{\sum_{t=1}^{T} \gamma_t(j) X_t}{\sum_{t=1}^{T} \gamma_t(j)},$$

$$(\sigma_j^2)^{new} = \frac{\sum_{t=1}^{T} \gamma_t(j)(X_t - \mu_j^{new})^2}{\sum_{t=1}^{T} \gamma_t(j)}.$$

Expectation-Maximization can converge to a local optimum rather than the global optimum. For that reason the project fits a deterministic set of starting seeds for each K, keeps converged finite-likelihood fits, chooses the greatest likelihood, and resolves a likelihood tie with the smallest seed (Baum et al. 164–171; Rabiner 257–286). State labels are then normalized by increasing fitted mean VIX change so that state identities are reproducible.

2.4 Viterbi path versus smoothed probabilities

The **Viterbi sequence** solves a joint decoding problem: it is the single most likely complete latent-state path conditional on the observed sequence (Viterbi 260–269; Rabiner 257–286). It is not the same object as selecting the largest posterior state probability independently at each date.

Smoothed state probabilities instead report $P(S_t = j \mid X_{1:T})$: each historical probability conditions on the **entire sample**, including observations after date t. They are valuable for retrospective description but are non-causal for historical dates. The fitted-state figures below therefore describe the full-sample regime structure; they are not real-time forecasts.

In [4]:
```
nb.step_2_hmm_diagnostics()
```

Gaussian HMM candidate: K=2

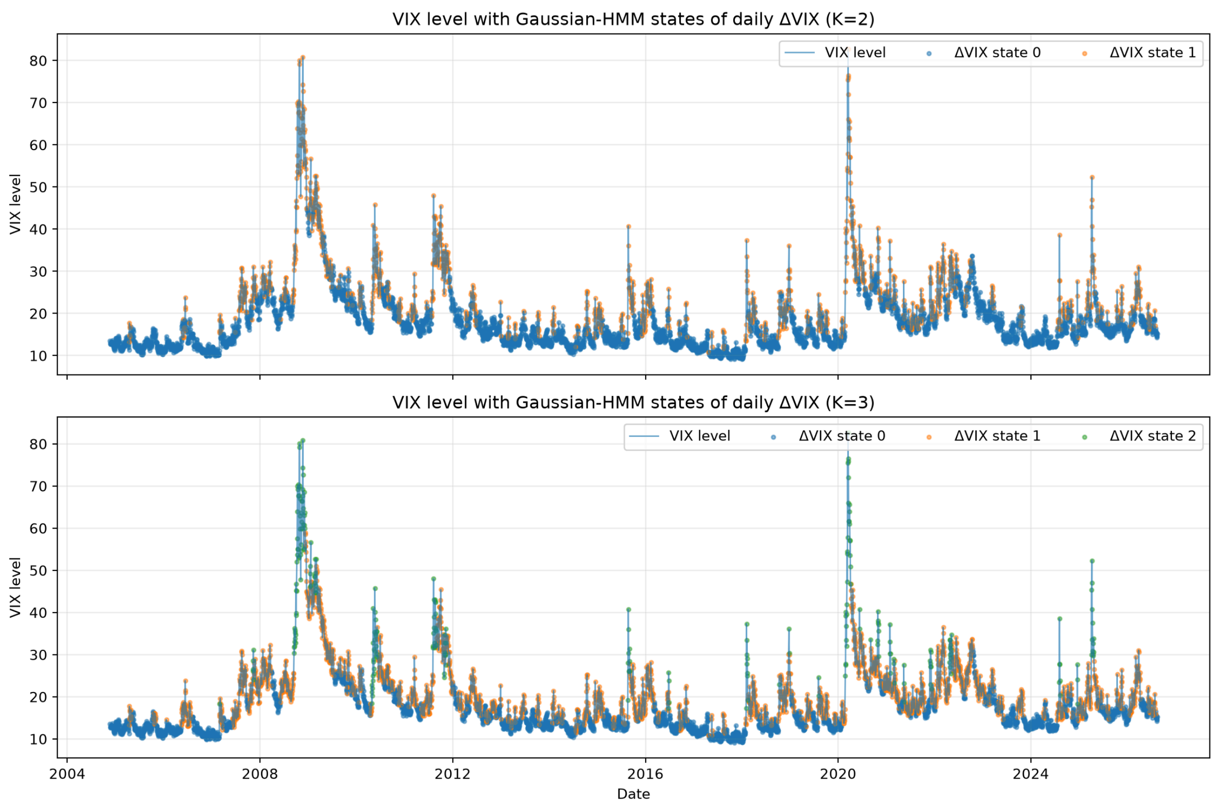
	state	mean_vix_change	variance_vix_change	start_probability	viterbi_obser
0	0	-0.074387	0.654348	1.000000e+00	
1	1	0.212640	11.957452	1.034965e-36	

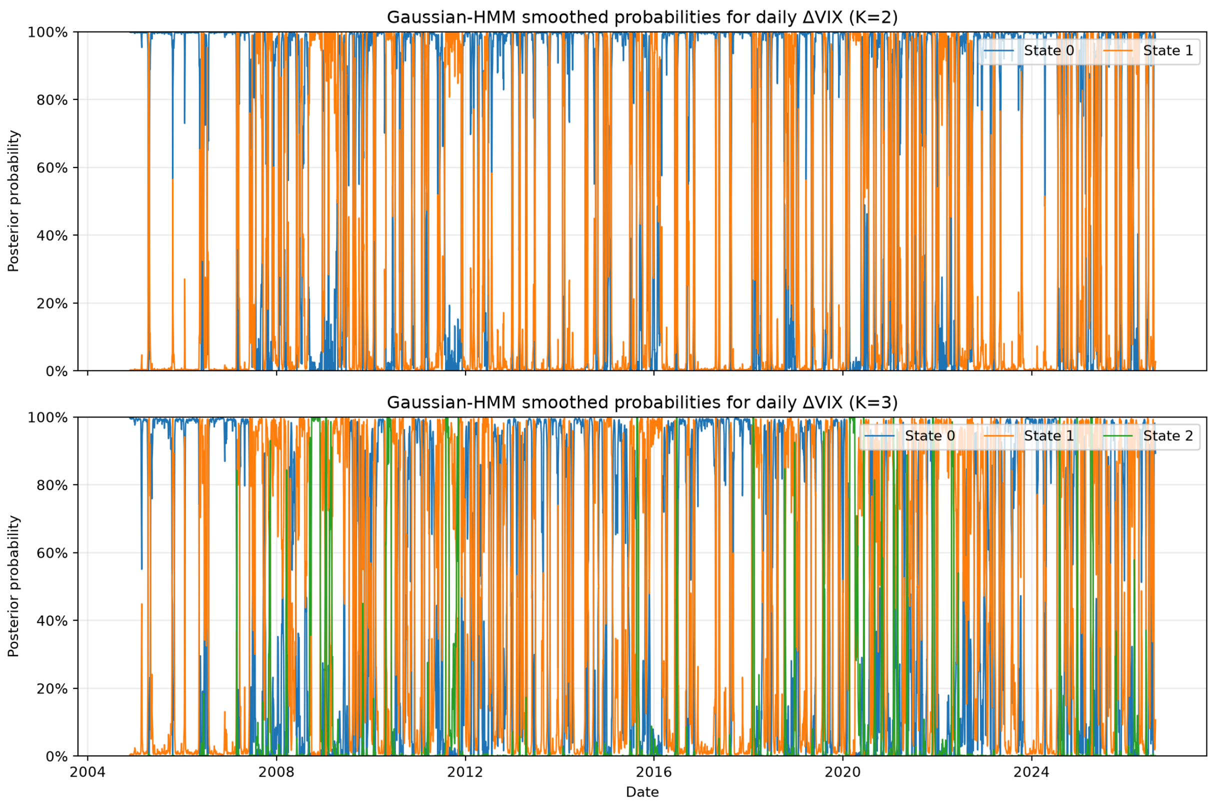
	from_state	state_0	state_1
0	0	0.958717	0.041283
1	1	0.117102	0.882898

Gaussian HMM candidate: K=3

	state	mean_vix_change	variance_vix_change	start_probability	viterbi_obser
0	0	-0.062142	0.384075	1.000000e+00	
1	1	-0.011439	3.118842	9.246767e-101	
2	2	0.659519	36.347398	3.003336e-208	

	from_state	state_0	state_1	state_2
0	0	9.388732e-01	0.059776	0.001351
1	1	7.568002e-02	0.908619	0.015701
2	2	1.193886e-23	0.128159	0.871841




Gaussian-HMM smoothed probabilities for daily ΔVIX (K=2)
Posterior probability
100%
80%
60%
40%
20%
0%
State 0
State 1
Gaussian-HMM smoothed probabilities for daily ΔVIX (K=3)
Posterior probability
100%
80%
60%
40%
20%
0%
State 0
State 1
State 2
2004
2008
2012
2016
2020
2024
Date

```
Out[4]: {2: {'parameters':    state  mean_vix_change  variance_vix_change  start_pr
        obability  \
          0      0        -0.074387             0.654348       1.000000e+00
          1      1         0.212640            11.957452       1.034965e-36

             viterbi_observations  viterbi_occupancy  posterior_mean_probability
          0                  4159           0.761025                    0.739429
          1                  1306           0.238975                    0.260571  ,
          'transition':    from_state   state_0   state_1
          0           0  0.958717  0.041283
          1           1  0.117102  0.882898,
          'states':             state
          Date
          2004-11-19      0
          2004-11-22      0
          2004-11-23      0
          2004-11-24      0
          2004-11-26      0
          ...           ...
          2026-08-11      0
          2026-08-12      0
          2026-08-13      0
          2026-08-14      0
          2026-08-17      0

          [5465 rows x 1 columns]},
         3: {'parameters':    state  mean_vix_change  variance_vix_change  start_pr
        obability  \
          0      0        -0.062142             0.384075       1.000000e+00
          1      1        -0.011439             3.118842      9.246767e-101
          2      2         0.659519            36.347398      3.003336e-208

             viterbi_observations  viterbi_occupancy  posterior_mean_probability
          0                  2900           0.530650                    0.521821
          1                  2306           0.421958                    0.421097
          2                   259           0.047392                    0.057082  ,
          'transition':    from_state       state_0   state_1   state_2
          0           0  9.388732e-01  0.059776  0.001351
          1           1  7.568002e-02  0.908619  0.015701
          2           2  1.193886e-23  0.128159  0.871841,
          'states':             state
          Date
          2004-11-19      0
          2004-11-22      0
          2004-11-23      0
          2004-11-24      0
          2004-11-26      0
          ...           ...
          2026-08-11      0
          2026-08-12      0
          2026-08-13      0
          2026-08-14      0
          2026-08-17      0

          [5465 rows x 1 columns]}}
```

Step 3 — HMM state-count selection

The comparison is restricted to the two HMM candidates. For K states, the project counts $K^2 + 2K - 1$ free parameters. A candidate is valid only when convergence, finite parameters and likelihood, probability normalization, Viterbi-label validity, and the project minimum state-occupancy rule all hold.

AIC penalizes likelihood by twice the parameter count (Akaike 716–723):

$$AIC = 2k - 2\log L.$$

BIC uses the sample-size-dependent penalty proposed by Schwarz (Schwarz 461–464):

$$BIC = k\log n - 2\log L.$$

The preferred model is the **valid HMM candidate with the smallest BIC**; a BIC tie within the fixed numerical tolerance selects the lower state count. The 5% minimum Viterbi occupancy threshold is a project stability rule, not a statistical theorem. If neither candidate is valid, the pipeline fails rather than falling back to another model family.

In [5]:
```
nb.step_3_hmm_selection()
```

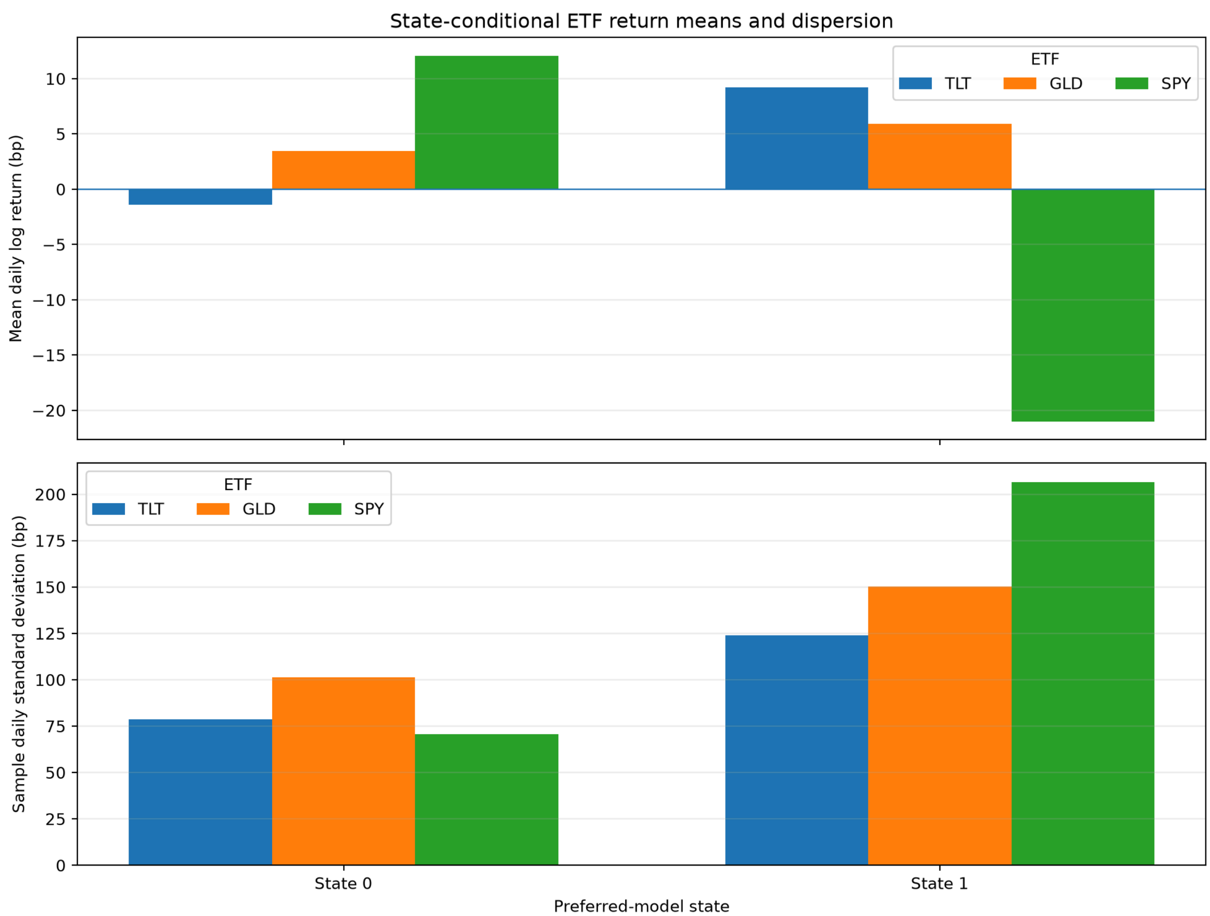
Step 3 — HMM-only model selection

	family	n_states	log_likelihood	n_parameters	n_observations	aic
0	hmm	2	-9306.004199	7	5465	18626.008398
1	hmm	3	-8959.534974	14	5465	17947.069948

	selected_model
family	hmm
input_data_sha256	c6b0e67a145f809313986ca0ef726d1dd42ffdaf57b04f...
n_states	2
selected_states_path	reports/tables/step3_selected_states.csv
selection_reason	Among valid HMM candidates, BIC selected K=2; ...
state_source	reports/tables/step2_hmm_2_states.csv

	state
Date	
2004-11-19	0
2004-11-22	0
2004-11-23	0
2004-11-24	0
2004-11-26	0

	state	asset	mean_log_return	std_log_return	observations
0	0	TLT	-0.000139	0.007855	4159
1	0	GLD	0.000347	0.010130	4159
2	0	SPY	0.001206	0.007058	4159
3	1	TLT	0.000920	0.012392	1306
4	1	GLD	0.000589	0.015030	1306
5	1	SPY	-0.002101	0.020646	1306



Out[5]:
```
{'family': 'hmm',
 'input_data_sha256': 'c6b0e67a145f809313986ca0ef726d1dd42ffdaf57b04f091848
ae1b20d261c1',
 'n_states': 2,
 'selected_states_path': 'reports/tables/step3_selected_states.csv',
 'selection_reason': 'Among valid HMM candidates, BIC selected K=2; a BIC t
ie within tolerance is resolved toward the lower state count.',
 'state_source': 'reports/tables/step2_hmm_2_states.csv'}
```

Step 4 — State-conditional ETF allocation

For every state of the selected HMM, TLT, GLD, and SPY are ranked by descending historical state-conditional mean daily log return. Exact equal means use the fixed priority **TLT → GLD → SPY**. Both assignment allocation methods use the same ranking:

- **100% Keep:** 100% to rank 1, 0% to ranks 2 and 3.
- **60/40 Spread:** 60% to rank 1, 40% to rank 2, 0% to rank 3.

The weights are long-only, finite, and sum to one. The ranking is an in-sample descriptive allocation rule, not a mean-variance optimizer. Concentrating on the historical top-ranked asset can increase estimation and concentration risk; diversification is therefore a relevant economic interpretation when comparing the two rules (Markowitz 77–91).

In [6]:
```
nb.step_4_dual_allocations()
```

Step 4 — 100% Keep

	method	state	rank_1_asset	rank_2_asset	rank_1_mean_log_return	rank_2_
0	100_keep	0	SPY	GLD	0.001206	
1	100_keep	1	TLT	GLD	0.000920	

Step 4 — 60/40 Spread

	method	state	rank_1_asset	rank_2_asset	rank_1_mean_log_return	ran
0	60_40_spread	0	SPY	GLD	0.001206	
1	60_40_spread	1	TLT	GLD	0.000920	

```
Out[6]: {'100_keep':      method  state rank_1_asset rank_2_asset  rank_1_mean_log_
return  \
 0  100_keep      0          SPY          GLD                0.001206
 1  100_keep      1          TLT          GLD                0.000920

    rank_2_mean_log_return  TLT_weight  GLD_weight  SPY_weight
 0                0.000347         0.0         0.0         1.0
 1                0.000589         1.0         0.0         0.0  ,
 '60_40_spread':          method  state rank_1_asset rank_2_asset  rank_1_m
ean_log_return  \
 0  60_40_spread      0          SPY          GLD                0.001206
 1  60_40_spread      1          TLT          GLD                0.000920

    rank_2_mean_log_return  TLT_weight  GLD_weight  SPY_weight
 0                0.000347         0.0         0.4         0.6
 1                0.000589         0.6         0.4         0.0  }
```

Step 5 — Backtest and benchmark comparison

5.1 Execution convention

ETF log returns are converted to simple returns before portfolio arithmetic:

$$r_{i,t}^{simple} = \exp(r_{i,t}^{log}) - 1.$$

For both HMM allocation methods, the state recorded on row $t-1$ determines the weights applied to ETF simple returns on row t. The first row is excluded. This one-row execution lag prevents same-row state execution.

The two required benchmarks are:

1. **Equal-weight monthly:** one-third TLT, one-third GLD, and one-third SPY at the start and at the first comparison date of each new calendar month, with weights drifting inside the month.
2. **SPY buy-and-hold:** the aligned SPY simple-return series.

All four portfolio return series use exactly the same dates.

5.2 Performance metrics

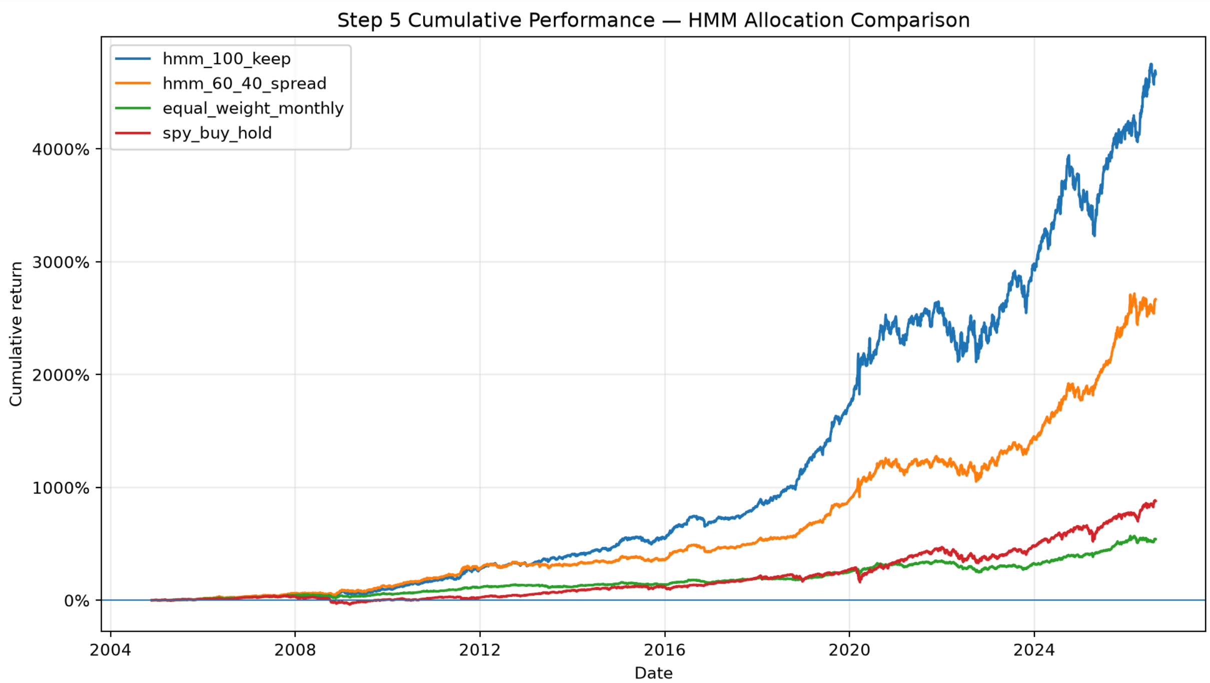
The evaluation reports exactly five metrics for each portfolio: cumulative return, annualized return, annualized volatility, Sharpe ratio, and maximum drawdown. Annualization uses 252 trading days, the Sharpe calculation uses a zero risk-free rate and sample standard deviation, and drawdown starts from initial wealth $W_0 = 1$.

The cumulative-performance figure is the primary visual comparison because it places both HMM allocation rules and both required benchmarks on the same wealth-growth scale. The numerical table should be read jointly with drawdown and volatility rather than as a return-only ranking.

In [7]:
```
nb.step_5_hmm_dual_method_comparison()
```

Step 5 — One-observation-lag portfolio comparison

	portfolio	cumulative_return	annualized_return	annualized_volatil
0	hmm_100_keep	46.629595	0.195044	0.142
1	hmm_60_40_spread	26.674627	0.165491	0.119
2	equal_weight_monthly	5.420849	0.089548	0.097
3	spy_buy_hold	8.798148	0.110994	0.188



```
Out[7]: {'daily':             hmm_100_keep  hmm_60_40_spread  equal_weight_monthly  
spy_buy_hold
 Date                                                                       
 2004-11-22      0.004769          0.004380              0.004592      0.00
4769
 2004-11-23      0.001526         -0.000864             -0.000560      0.00
1526
 2004-11-24      0.002370          0.004103              0.003016      0.00
2370
 2004-11-26     -0.000760          0.001675             -0.000655     -0.00
0760
 2004-11-29     -0.004563         -0.001766             -0.004237     -0.00
4563
 ...                  ...               ...                   ...          
...
 2026-08-11     -0.003195         -0.003487             -0.001914     -0.00
3195
 2026-08-12      0.002505          0.005453              0.003915      0.00
2505
 2026-08-13      0.006977         -0.001701             -0.000873      0.00
6977
 2026-08-14     -0.001980          0.001339             -0.000664     -0.00
1980
 2026-08-17     -0.004727          0.001159             -0.000864     -0.00
4727

 [5464 rows x 4 columns],
 'summary':               portfolio  cumulative_return  annualized_return  
\
 0          hmm_100_keep          46.629595           0.195044   
 1      hmm_60_40_spread          26.674627           0.165491   
 2  equal_weight_monthly           5.420849           0.089548   
 3          spy_buy_hold           8.798148           0.110994   

    annualized_volatility  sharpe_ratio  max_drawdown  observations  
 0               0.142125      1.325171     -0.195403          5464  
 1               0.119317      1.343538     -0.164250          5464  
 2               0.097112      0.931856     -0.230437          5464  
 3               0.188845      0.651956     -0.551894          5464  }
```

5.3 State-count × allocation-method sensitivity

The sensitivity analysis crosses HMM state counts $K = 2$ and $K = 3$ with both allocation methods, producing exactly four rows. It reuses the persisted HMM candidate state paths, the common lagged return-date intersection, the same state-statistics/ranking logic, the same backtest engine, and the same performance metrics.

Sensitivity is diagnostic rather than a second model-selection rule: a $K = 3$ row may be shown even if that candidate fails the 5% main-selection occupancy requirement. The selected Step 3 model remains unchanged.

In [8]:
```
sensitivity_nb.step_5_state_count_sensitivity()
```

Step 5 — HMM state-count and allocation-method sensitivity

	family	n_states	method	cumulative_return	annualized_return	annuali
0	hmm	2	100_keep	46.629595	0.195044	
1	hmm	2	60_40_spread	26.674627	0.165491	
2	hmm	3	100_keep	35.894722	0.181051	
3	hmm	3	60_40_spread	20.153970	0.151137	

Out[8]:

	family	n_states	method	cumulative_return	annualized_return	annua
0	hmm	2	100_keep	46.629595	0.195044	
1	hmm	2	60_40_spread	26.674627	0.165491	
2	hmm	3	100_keep	35.894722	0.181051	
3	hmm	3	60_40_spread	20.153970	0.151137	

Interpretation, limitations, and conclusions

Look-ahead limitation

The one-row execution lag removes same-row execution, but it **does not make this full-sample exercise causal**. Three core components use information unavailable at earlier historical dates:

- HMM parameters are estimated on the full sample.
- Viterbi decoding and smoothed posterior probabilities use the full observation sequence.
- State-conditional ETF rankings are computed from full-sample state assignments and returns.

Consequently, the reported results are **assignment backtest results**, not evidence of a deployable live trading edge. A production study would need expanding or rolling re-estimation, one-sided state inference, training-only asset ranking, and a strictly held-out evaluation period.

Statistical interpretation

Trying alternative state counts and allocation rules creates model-selection and data-snooping risk. Backtest performance can be overstated when the same data influence specification choices and evaluation (White 1097–1126). Sharpe ratios are also vulnerable to selection bias, backtest overfitting, and non-normal return distributions when many alternatives are investigated (Bailey and Lopez de Prado 94–107). The four-row sensitivity table is therefore evidence about robustness within the assignment design, not a multiple-testing correction.

Conclusion

This revision isolates the analysis to one coherent regime framework: Gaussian HMMs on VIX changes. It distinguishes estimation, joint-path decoding, and smoothed posterior inference; selects only between $K = 2$ and $K = 3$ HMM candidates under fixed validity and BIC rules; and evaluates both required allocation mappings against identical-date benchmarks. The central empirical comparison is intentionally transparent: **HMM 100% Keep**, **HMM 60/40 Spread**, **monthly equal weight**, and **SPY buy-and-hold**, with the same five performance metrics and a state-count × allocation sensitivity check.

The analysis should therefore be interpreted as a reproducible retrospective study of regime-conditioned allocation mechanics. Its strongest methodological result is not a claim of forecastability, but a clear separation between what the HMM estimates, how state information is decoded, how allocations are constructed, and where full-sample information creates look-ahead bias.

Works Cited — MLA 9

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The machine-readable bibliography registry used by the repository is exposed below.

In [9]:
```
nb.canonical_works_cited()
```

Bibliography registry: `reports/references.bib`